# Database Design Summary

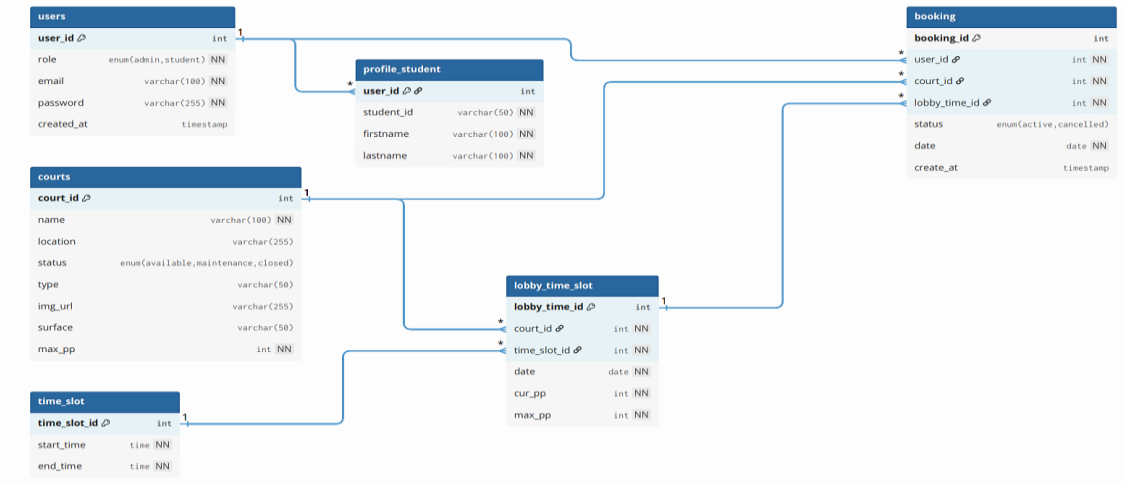
System: Court Booking System
Databases Used: MySQL (Relational) and MongoDB (NoSQL)

## 1. MySQL Database Schema

MySQL is used as the primary relational database to store structured data such as users, courts, time slots, and bookings.

| Table | Description |
|---|---|
| **user** | Stores user account information (email, password, role) |
| **profile_student** | Stores student profile details |
| **courts** | Stores court information such as name, type, and location |
| **time_slot** | Defines available start and end times |
| **lobby_time_slot** | Defines which time slots are available for each court and date |
| **booking** | Stores booking records created by users |

ER Diagram:




## 2. MongoDB Document Structure

**field_reviews**

```
{
  _id: ObjectId,
  court_id: Number,
  court_name: String,
  user_id: Number,
  rating: Number,
  comment: String,
  created_at: Date
}
```

**booking_logs**

```
{
  _id: ObjectId,
  action: String,
  user_id: Number,
  detail: { booking_id, court_id, date, time_slot_id },
  status: String,
  ip_address: String,
  created_at: Date
}
```

**auth_logs**

```
{
  _id: ObjectId,
  action: String,
  detail: { email, reason },
  status: String,
  ip_address: String,
  created_at: Date
}
```

## 3. Role of Each Database in the System

MySQL: Stores core system data such as users, courts, time slots, and booking list.
MongoDB: Stores logs and user-generated content including booking activity logs, authentication logs, and court reviews.